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Data Communications

Internetworking



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Internetworking – Network layer

Introduction

- The network layer is a part of the communication process in computer networks.
- Its main job is to move data packets between different networks.
- It helps route these packets from the sender to the receiver across multiple paths and networks.
- The Internet Protocol (IP) is a key protocol used at this layer, along with other protocols for routing, testing, and encryption.

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Internetworking – Network layer

Features of Network Layer

- The main responsibility of the Network layer is to carry the data packets from the source to the destination without changing or using them.
- If the packets are too large for delivery, they are **fragmented** i.e., broken down into smaller packets.
- It decides the route to be taken by the packets to travel from the source to the destination among the multiple routes available in a network (also called **routing**).

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Internetworking – Network layer

Services Offered by Network Layer

The **services** which are offered by the network layer protocol are as follows:

- Packetizing
- Routing
- Forwarding

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Packetizing

- The process of encapsulating the data received from the upper layers of the network (also called payload) in a network layer packet at the source and decapsulating the payload from the network layer packet at the destination is known as packetizing.
- The source host adds a header that contains the source and destination address and some other relevant information required by the network layer protocol to the payload received from the upper layer protocol and delivers the packet to the data link layer.

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Routing

- Routing is the process of moving data from one device to another device.
- In a network, there are a number of routes available from the source to the destination. The network layer specifies some strategies which find out the best possible route. This process is referred to as routing.

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Forwarding

- Forwarding is simply defined as the action applied by each router when a packet arrives at one of its interfaces.
- When a router receives a packet from one of its attached networks, it needs to forward the packet to another attached network (unicast routing) or to some attached networks (in the case of multicast routing).
- Routers are used on the network for forwarding a packet from the local network to the remote network. So, **the process of routing involves packet forwarding from an entry interface out to an exit interface.**

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Internetworking – Network layer



Questions:

1. Explain the key responsibilities of the Network Layer in the OSI model. How does it ensure end-to-end delivery of packets?
2. What is packetizing? Describe the process of encapsulation at the sender and decapsulation at the receiver.
3. Differentiate between routing and forwarding. Why are both necessary for internetworking?

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Principles of Internetworking



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Principles of Internetworking

Introduction

- Internetworking refers to connecting multiple computer networks together to function as a single, large network.
- The **principles of internetworking** are the fundamental rules and concepts that guide the design, operation, and communication across these interconnected networks

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Principles of Internetworking

- Transparency
- Scalability
- Interoperability
- Flexibility and Heterogeneity
- Efficiency
- Reliability and Fault Tolerance
- Security
- Routing and Addressing
- Quality of Service (QoS)

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Principles of Internetworking



Transparency

The internetwork should hide the complexity of underlying networks from users and applications.

Scalability

The internetwork must support growth (more users, devices, and data) without the performance degradation.

Interoperability

Different networks, hardware, and software must work together. This is achieved by **standard protocols** (like TCP/IP, OSI). Ex: A Windows system communicating with a Linux server using TCP/IP.

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Principles of Internetworking



Flexibility and Heterogeneity

They should work regardless of differences in hardware, topology, or transmission technology. Internetworks must support different types of networks:

- LANs (Ethernet, Wi-Fi)

- WANs (MPLS, leased lines)

- MANs (Metropolitan networks)

Efficiency

The design should minimize **latency, bandwidth consumption, and routing overhead.**

Includes - Optimal routing, Congestion control

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Principles of Internetworking



Reliability and Fault Tolerance

Internetworks must continue functioning even if some components fail. This is achieved using:

- Redundant paths (multiple routes)
- Error detection & correction (CRC, checksums)

Security

Protects data, users, and networks. It includes:

- Authentication (proving identity)
- Encryption (protecting data in transit)

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Principles of Internetworking

Routing and Addressing

- Addressing: Every device must have a unique identifier (IP address).
- Routing: Routers select the best path for data packets to travel.
- Protocols: RIP, OSPF, BGP.

Quality of Service (QoS)

Ensures that critical applications (like video calls, VoIP) get priority over less important traffic. Uses techniques like traffic shaping, prioritization, and bandwidth reservation.

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Principles of Internetworking



Questions:

- 1. Explain the principle of transparency in internetworking**
- 2. Discuss the importance of security in internetworking**
- 3. Discuss the role of routing and addressing in internetworking**

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Devices in Internetworking



Repeater

- Extends the range of a network by amplifying (regenerating) signals.
- Works at the Physical Layer (Layer 1).

Hub

- Connects multiple devices in a LAN and forwards data to all connected devices.
- Works at Physical Layer (Layer 1).
- Limitation: Broadcasts to all → causes collisions, inefficient.

Bridge

- Connects two LAN segments and filters traffic based on **MAC addresses**.
- Works at Data Link Layer (Layer 2).
- Benefit: Reduces collisions by segmenting networks

Switch

- Similar to a bridge but more powerful, forwards data to the specific device using MAC address table.
 - Works at Data Link Layer (Layer 2), some switches also work at Network Layer
- Benefit: Efficient, reduces collisions, improves performance.

Router

- Connects multiple networks (LAN to WAN, LAN to Internet), routes data based on IP addresses.
- Works at Network Layer (Layer 3).
- Benefit: Provides routing, traffic management, and path selection

Gateway

- Connects networks that use different protocols (e.g., TCP/IP to IPX/SPX).
- Works at all 7 OSI layers depending on implementation.
- Benefit: Provides interoperability between heterogeneous networks.

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Devices in Internetworking



Modem

- Converts **digital signals to analog** (and vice versa) for transmission over telephone lines.
- Works at Physical Layer. Use: Used in DSL, broadband connections

Access Point (AP)

- Provides wireless connectivity, allowing Wi-Fi devices to connect to a wired LAN.
- Used at Data Link Layer.

Firewall

- Monitors and controls incoming/outgoing network traffic based on security rules.
- Works at **Network Layer and above**.
- **Use:** Provides **security** by blocking unauthorized access.

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Internet Protocol



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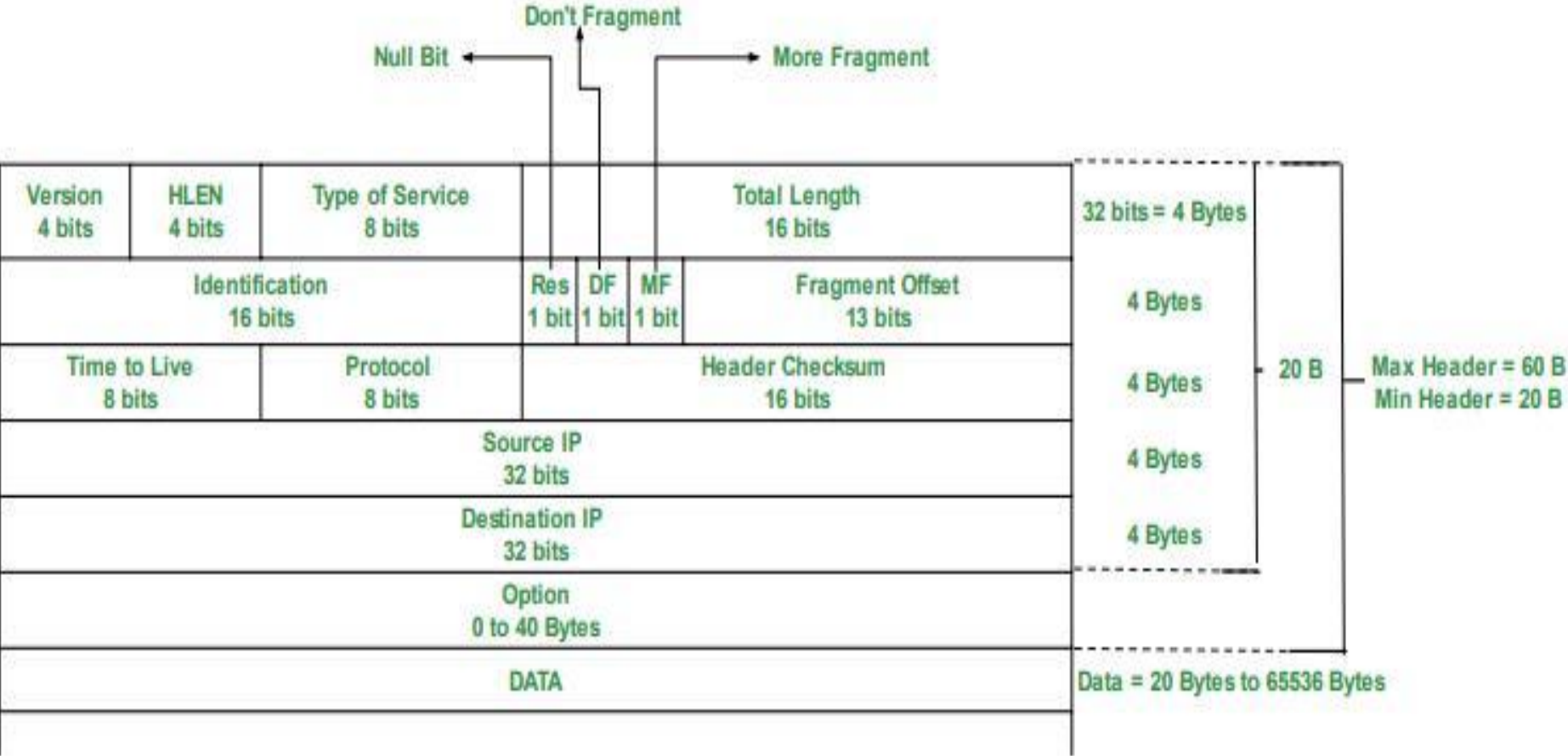
Internet Protocol

- IPv4 is a connectionless protocol.
- IPV4 is the fourth revision of the Internet Protocol
- It is widely used in data communication over different kinds of networks.
- IPv4 uses 32-bit addresses for communication in five classes: A, B, C, D and E.
- IPv4 uses 32-bit (4-byte) addressing, which gives 2^{32} addresses.
- IPv4 addresses are written in the dotted-decimal notation, which comprises four octets of the address expressed individually in decimal and separated by periods.

Example: 192.168.1.5

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IP - Header



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Internet Protocol - Datagram Header

- **VERSION:** Version of the IP protocol (4 bits), which is 4 for IPv4
- **HLEN:** IP header length (4 bits), which is the number of 32 bit words in the header. The minimum value for this field is 5.
- **Type of service:** Low Delay, High Throughput, Reliability (8 bits)
- **Total Length:** Length of header + Data (16 bits), which has a minimum value 20 bytes and the maximum is 65,535 bytes.
- **Identification:** Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits)
- **Flags:** 3 flags of 1 bit each : reserved bit (must be zero), do not fragment flag, more fragments flag (same order)

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Internet Protocol



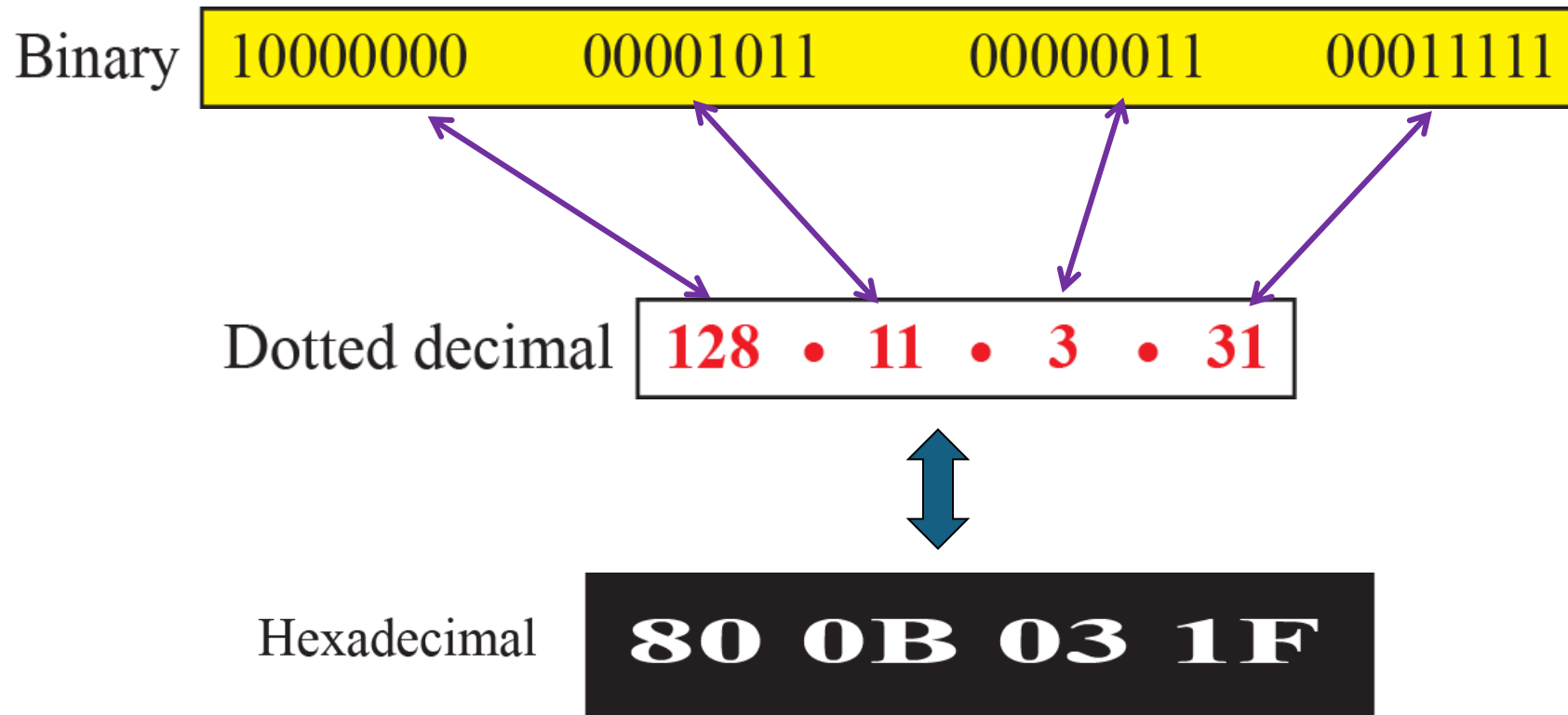
- **Fragment Offset:** Indicates where in the original datagram this fragment belongs (13 bits)
- **Time to live(TTL):** Datagram's lifetime (8 bits), It prevents the datagram to loop through the network by restricting the number of Hops.
- **Protocol:** Name of the protocol to which the data is to be passed (8 bits).

For example: TCP =6, UDP = 17, ICMP =1

- **Header Checksum:** 16 bits header checksum for checking errors in the datagram header
- **Source IP address:** 32 bits IP address of the sender
- **Destination IP address:** 32 bits IP address of the receiver
- **Option:** Optional information such as source route, record route. Used by the Network administrator to check whether a path is working or not.

- Three different notations in IPv4 addressing

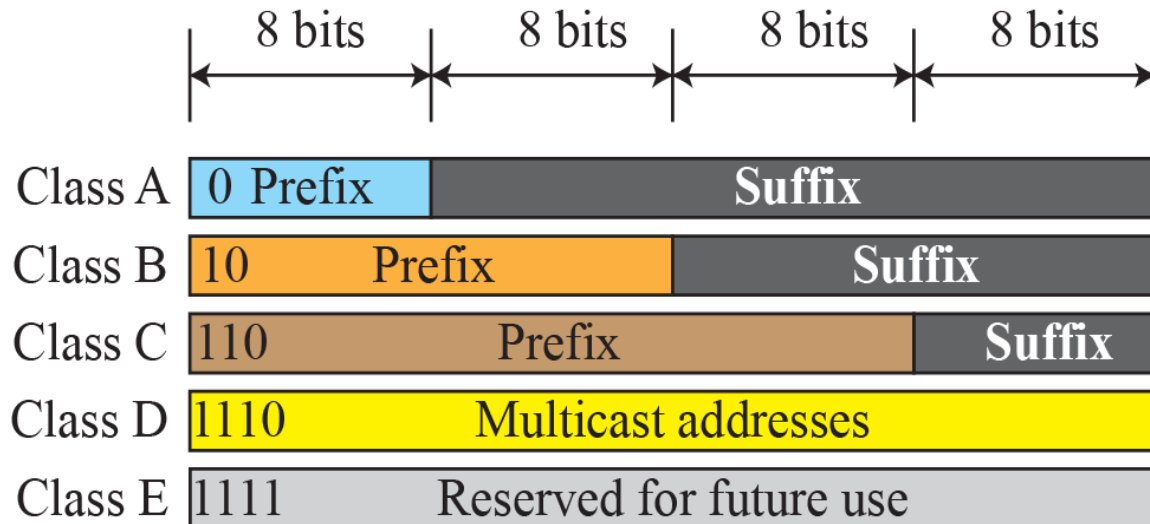
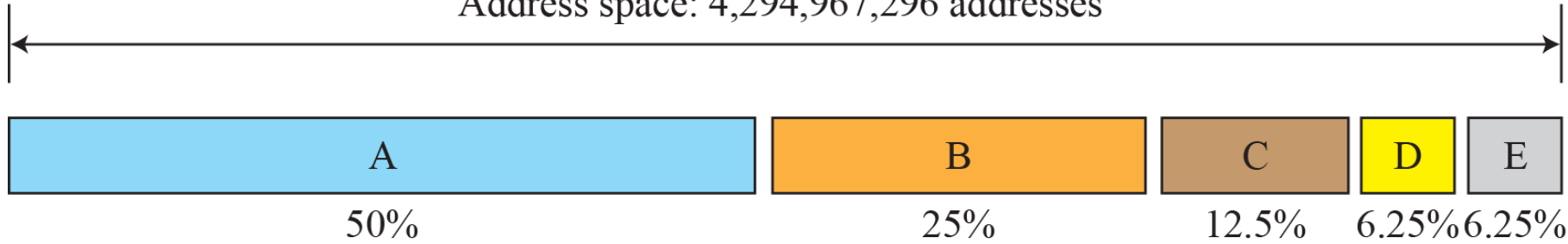
Address space – 2^{32}



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IPV4 Addresses

Address space: 4,294,967,296 addresses



Class	Prefixes	First byte
A	$n = 8$ bits	0 to 127
B	$n = 16$ bits	128 to 191
C	$n = 24$ bits	192 to 223
D	Not applicable	224 to 239
E	Not applicable	240 to 255

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IP Addresses - Numericals



Q1: IP Address: 192.168.1.10 Represent it in binary

Convert each octet to binary:

192 - 11000000

168 - 10101000

1 - 00000001

10 - 00001010

Answer:

192.168.1.10 - 11000000.10101000.00000001.00001010

Problem 2: Convert IP from Binary to Decimal

IP Address: 11000000.10101000.00000001.00001010

Convert each octet to decimal:

- 11000000 - $128+64 = 192$
- 10101000 - $128+32+8 = 168$
- 00000001 - 1
- 00001010 - $8+2 = 10$

Answer:

11000000.10101000.00000001.00001010 - 192.168.1.10

Q1. Represent IP Address: 172.16.5.25 in binary

Q2. Represent IP Address: 11000000.10101000.00001010.00001111 in decimal

Q3. convert the IP Address: 172.31.255.1 to binary

Q4. Convert the IP Address: 00001010.00010000.00000001.00000110 to decimal

Public IP Address: This address is available publicly and is assigned by the network provider to the router. Public IP Addresses are of two types:

- 1. Dynamic IP address**
- 2. Static IP address**

Dynamic IP Address: When you connect a smartphone or computer to the internet, the ISP provides you an IP Address from the range of available IP Addresses.

As, the IP Address keeps on changing every time when you connect to the internet, it is called a Dynamic IP Address.

Example:

- The laptop connects to Wi-Fi today and gets 192.168.1.12, tomorrow it may get 192.168.1.15.
- A home broadband user is given a different public IP ex: 49.207.123.10 today, and 49.207.88.65 tomorrow by the ISP

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IP Addresses - classification



Static IP Address:

- Static address never changes.
- They serve as a permanent internet address.
- These are used by DNS servers.
- Static IP Address provides information such as device is located on which continent, which country, which city, and which Internet Service Provider provides internet connection to that particular device.

Example:

- A company's web server has a fixed public IP: 203.0.113.25, so users worldwide can access it.
- A printer in a school lab always uses 192.168.1.50, so all computers can find it easily.
- Google DNS server 8.8.8.8 (always the same)

Private IP Address:

- This is an internal address of your device
- A Private IP is used inside a local network (LAN).
- It is not routable on the internet (cannot be accessed directly from outside).
- Helps devices within the same network (like computers, printers, mobiles) to communicate.
- These addresses are defined by IANA in specific ranges.

Private IP Address Ranges

- **Class A** → 10.0.0.0 to 10.255.255.255
- **Class B** → 172.16.0.0 to 172.31.255.255
- **Class C** → 192.168.0.0 to 192.168.255.255

- A subnet mask is a 32 bit number, that divides an IP address into two parts:

Network part - identifies the network.

Host part - identifies the device (computer, printer, etc.) inside that network.

- It tells, which portion of the IP address belongs to the network and which portion belongs to the host.
- The natural masks or default subnet masks of class a, b &c are:

Class a: 255.0.0.0

Class b: 255.255.0.0

Class c: 255.255.255.0

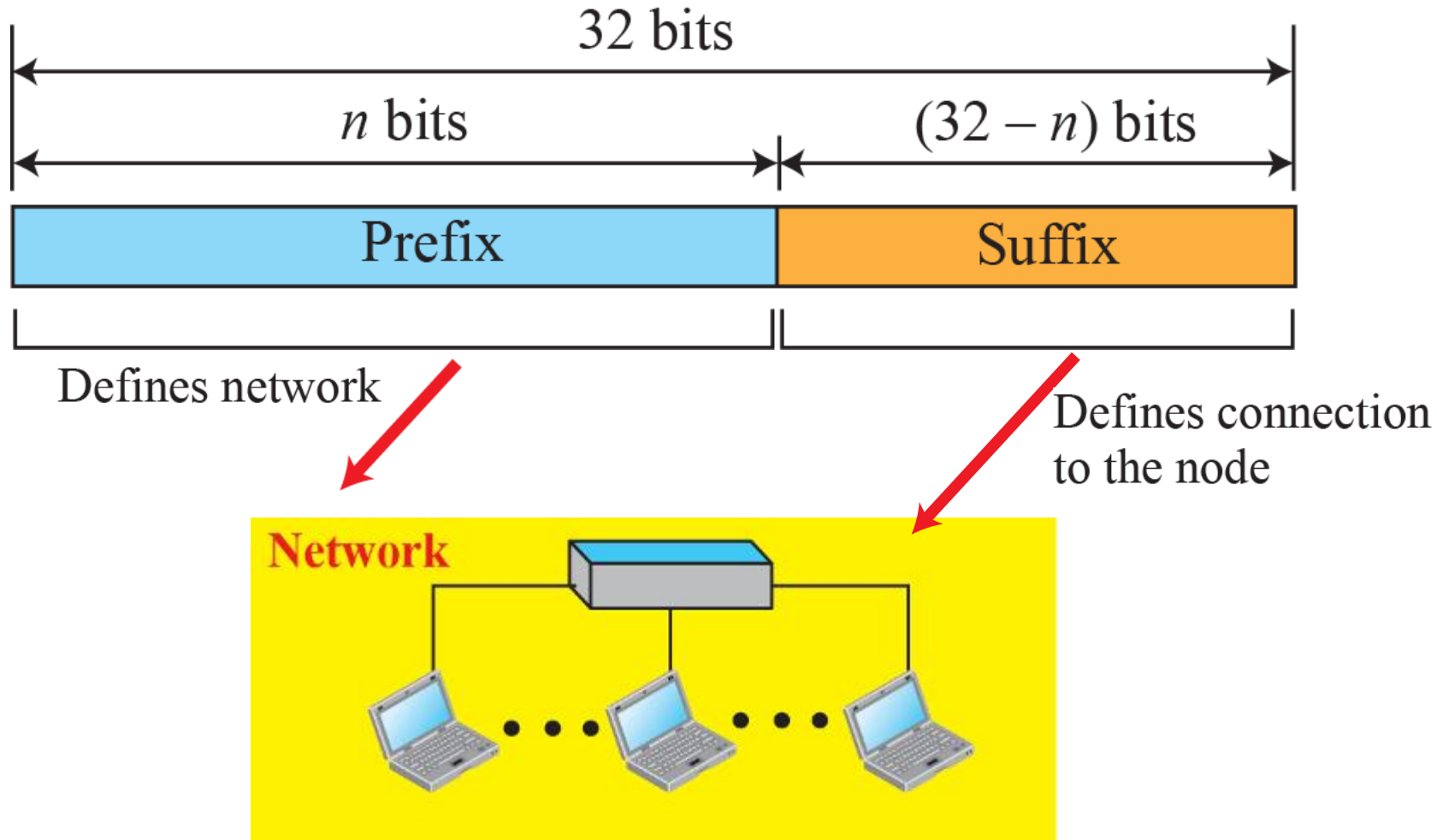
Example

IP Address: **192.168.1.10**, Subnet Mask: **255.255.255.0**

Then **192.168.1** = Network part, **.10** = Host part (device in that network)

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Hierarchy in addressing



Find the Network Address

IP Address: 192.168.1.10 and **Subnet Mask:** 255.255.255.0

Steps:

- Convert both to binary:
IP - 11000000.10101000.00000001.00001010
Mask - 11111111.11111111.11111111.00000000
- **Perform AND operation** (bit by bit):
 - Result - 11000000.10101000.00000001.00000000
- Convert back to decimal - 192.168.1.0

Answer:

Network Address = **192.168.1.0**

Problem: Find the Broadcast Address for the given

IP Address: 192.168.1.10 and Subnet Mask: 255.255.255.0

Steps:

- Network part = 192.168.1
- Host part - all 1s, so it is - 255

Answer: Broadcast Address = 192.168.1.255

Find Number of Hosts for the **IP:** 192.168.10.0 and **Subnet Mask:** 255.255.255.240

Solution:

Number of hosts = $2^{\text{number of host bits}} - 2$

Host bits = 32 – number of 1s in mask

255.255.255.240: 28 network bits, 4 host bits

Number of hosts = $2^4 - 2 = 14$

Answer: 14 hosts per subnet

Q: Calculate the number of hosts for the IP: 192.168.1.0 and Subnet Mask: 255.255.255.240

Solution:

Convert subnet mask to binary - 255.255.255.240 = 11111111.11111111.11111111.11110000

Count **0s in the mask** - 4 host bits

Apply formula : $2^4 - 2 = 16 - 2 = 14$

Answer: 14 hosts

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Subnet Mask



Find Network and Broadcast Address

IP: 192.168.20.25 and **Subnet Mask:** 255.255.255.192

Solution:

Subnet mask in binary - 255.255.255.192 = 11111111.11111111.11111111.11000000

Network part: keep network bits same: 192.168.20.0 (nearest multiple of 64 in last octet)

Broadcast address: set host bits to 1: 192.168.20.63

Answer:

Network Address = 192.168.20.0

Broadcast Address = 192.168.20.63

Find Number of Hosts

IP: 172.16.8.0 and **Subnet Mask:** 255.255.255.224

Solution:

Convert subnet mask $\rightarrow 255.255.255.224 = 11111111.11111111.11111111.11100000$

Host bits = 5 (number of 0s)

Apply formula $\rightarrow 2^5 - 2 = 32 - 2 = 30$

Answer:

Host Bits: 5

Number of Hosts: 30

Network Address: 172.16.8.0

Broadcast Address: 172.16.8.31

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Subnet Mask



IP Address: 192.168.1.10 and **Subnet Mask:** 255.255.255.0.

Find the Network and broadcast address

Solution:

Network part → 192.168.1

Host part → .10

Network Address: 192.168.1.0

Broadcast Address: 192.168.1.255

- Q1: Find Network Address for the IP: 10.1.2.15 and Subnet Mask: 255.255.255.0
- Q2: Find Broadcast Address IP: 192.168.5.20 and Subnet Mask: 255.255.255.0
- Q3: Find Network Address for the IP: 172.16.12.34 and Subnet Mask: 255.255.255.0

Data Communications

CIDR- Classless Inter-Domain Routing



- CIDR stands for **Classless Inter-Domain Routing**.
- It is a method for allocating IP addresses and routing Internet Protocol packets more efficiently.
- Introduced in **1993** by the Internet Engineering Task Force (IETF).
- It replace the old **classful IP addressing system (Class A, B, C)**.
- It helps to **efficiently allocate IP addresses** and **reduce the size of routing tables**.
- It helps to divide or combine IP address ranges **based on need**, not fixed classes
- Format: **IP address / Prefix length**
 - Example: 192.168.1.0/24

CIDR notation = <Network Address>/<Prefix Length>

Example:

CIDR Notation	Subnet Mask	No. of Hosts	Network Type
192.168.1.0/24	255.255.255.0	254	Small network
10.0.0.0/8	255.0.0.0	16,777,214	Very large network
172.16.0.0/16	255.255.0.0	65,534	Medium network

Prefix length represents the number of bits used for **network** identification.



CIDR

Converting CIDR to Subnet Mask

Content:

Each bit of the prefix corresponds to 1 in the subnet mask.

CIDR	Subnet Mask	Binary Mask
/8	255.0.0.0	11111111.00000000.00000000.00000000
/16	255.255.0.0	11111111.11111111.00000000.00000000
/24	255.255.255.0	11111111.11111111.11111111.00000000
/26	255.255.255.192	11111111.11111111.11111111.11000000

Calculating Number of Hosts

Formula:

$$\text{Number of Hosts} = 2^{(32-\text{prefix})} - 2$$

Example:

For /24:

$$= 2^{(32-24)} - 2 = 256 - 2 = 254$$

Problem 1: Calculate Number of Hosts

How many hosts are there in the network 192.168.5.0/26?

Solution:

Formula:

$$\text{Number of hosts} = 2^{(32-\text{prefix})} - 2$$

Here, prefix = 26

$$2^{(32-26)} - 2 = 2^6 - 2 = 64 - 2 = 62$$

Answer: 62 hosts

Problem 2: Find the Subnet Mask

What is the subnet mask of 10.0.0.0/12?

Solution:

/12 → first 12 bits are network bits

Subnet mask: 255.240.0.0

Answer: 255.240.0.0

Problem 3: Determine Network Range

Find the range of IP addresses for 192.168.1.0/28

Solution:

/28 $\rightarrow 32 - 28 = 4$ bits for hosts $\rightarrow 2^4 = 16$ addresses

First IP (network) = 192.168.1.0

Last IP (broadcast) = 192.168.1.15

Usable IPs: 192.168.1.1 $\rightarrow$ 192.168.1.14

Advantages of CIDR

- Better utilization of IP address space
- Easier route aggregation
- Reduces routing table size
- Supports hierarchical IP allocation
- Enables flexible subnetting (VLSM)

CIDR - Questions



1. CIDR stands for _____.

Sol: Classless Inter-Domain Routing

2. What does /24 mean in CIDR notation?

Sol: First 24 bits are network bits

3. Number of hosts in 10.0.0.0/26?

Sol: 62

4. CIDR helps in reducing _____.

Sol: Routing table size

5. 192.168.0.0/22 represents how many IP addresses?

Sol: 1024

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Virtual Private Networks & IP Security



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Virtual Private Networks

- A virtual private network (VPN) is a technology that creates a safe and encrypted connection over a less secure network, such as the Internet.
- A VPN is a way to extend a private network using a public network such as the Internet.
- The name only suggests that it is a “Virtual Private Network”, i.e. user can be part of a local network sitting at a remote location.
- It makes use of tunneling protocols to establish a secure connection.

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Virtual Private Networks

Need for VPN

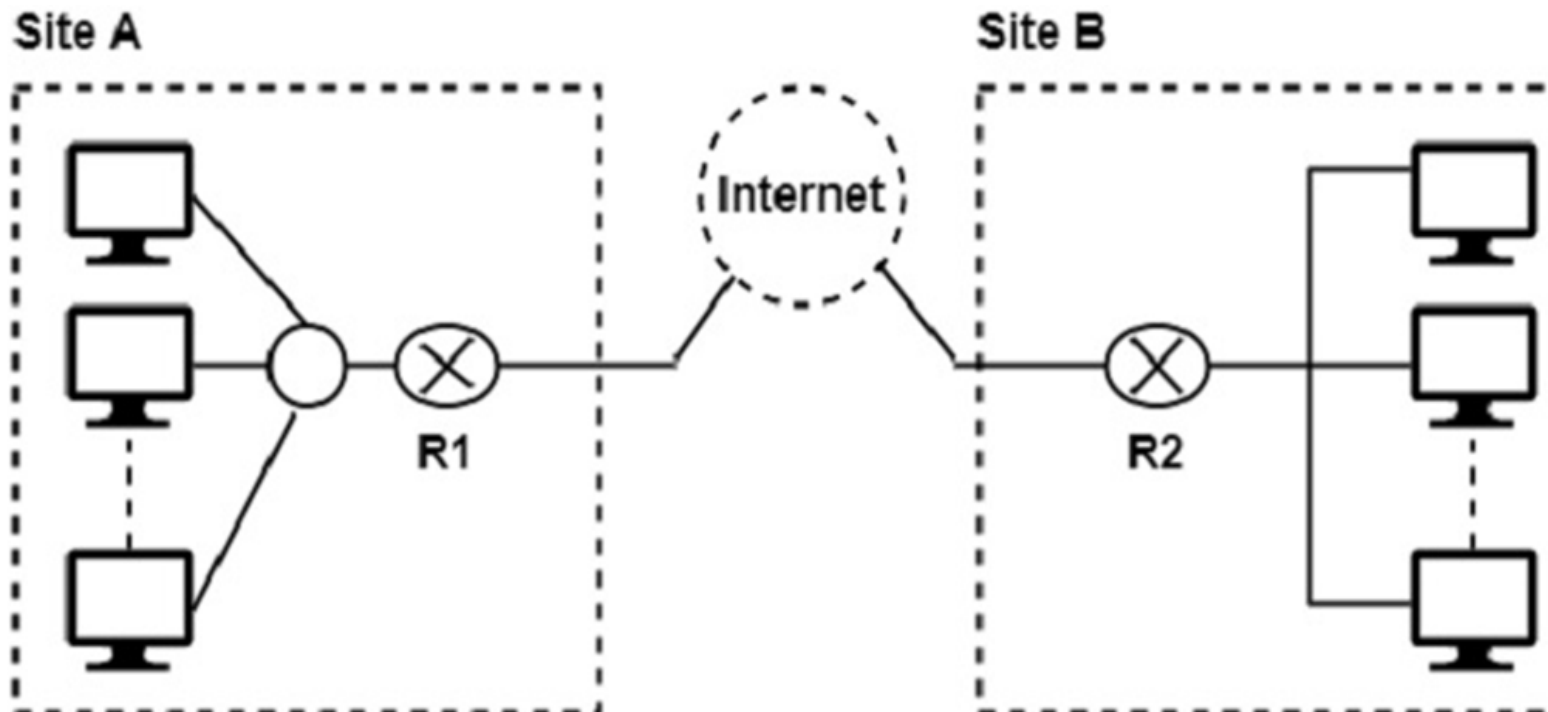
- VPNs are a necessity since privacy, security, and free internet access should be everybody's right.
- First, they establish secure access to the corporate networks for remote users.
- Secure the data during the transmission
- Help users to avoid geo-blocking and censorship.
- It prevents unauthorized people from eavesdropping on the traffic and allows the user to conduct work remotely
- VPNs are highly useful for protecting data on open Wi-Fi, for privacy

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Virtual Private Networks

VPN technology is widely used in corporate environments.

A VPN connection is shown in the figure below – Routers R1 and R2 use VPN technology to guarantee privacy for the organization



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Virtual Private Networks

VPN connections are used in two important ways –

- To establish WAN connections using VPN technology between two distant networks that may be thousands of miles apart, but where each has some way of accessing the internet.
- To establish remote access connections that enable remote users to access a private network through a public network like the internet.

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Virtual Private Networks

What does a VPN primarily protect against?

- A) Physical theft of the device
- B) Unauthorized access and eavesdropping over public networks
- C) Slow internet connections
- D) Power surges

Answer: B) Unauthorized access and eavesdropping over public networks

What is one major advantage of using a VPN?

- A) It guarantees 100% internet speed
- B) It hides your IP address and encrypts data
- C) It removes all malware from your computer
- D) It automatically updates your software

Answer: B) It hides your IP address and encrypts data

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Virtual Private Networks

Which of the following is NOT a type of VPN?

- A) Remote Access VPN
- B) Site-to-Site VPN
- C) Personal VPN
- D) Local Area VPN

Answer: D) Local Area VPN

What is the primary purpose of a VPN?

- A) To increase internet speed
- B) To provide secure communication over a public network
- C) To block all network traffic
- D) To replace the need for antivirus software

Answer: B) To provide secure communication over a public network

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IP Security



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IP Security



Definition and Purpose of IPSec

- IPSec is a set of protocols that provide security services for data transmitted over the internet or other public networks.
- By using cryptographic techniques, IPSec's main goal is to guarantee the ***confidentiality, integrity, and authenticity*** of data transmitted between parties in a network.

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IP Security - Components

- IPSec has different components that work together to provide secure communication:
 - **Authentication Header (AH),**
 - **Encapsulating Security Payload (ESP), and**
 - **Security Associations (SA).**
- AH is responsible for authenticating the source of data by adding a header to each IP packet,
- ESP encrypts data using symmetric encryption algorithms like AES or 3DES (**AES - Advanced Encryption Standard, DES - Data Encryption Standard**)
- SA manages negotiations between communicating devices in securing an encrypted connection.

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IP Security Scenario

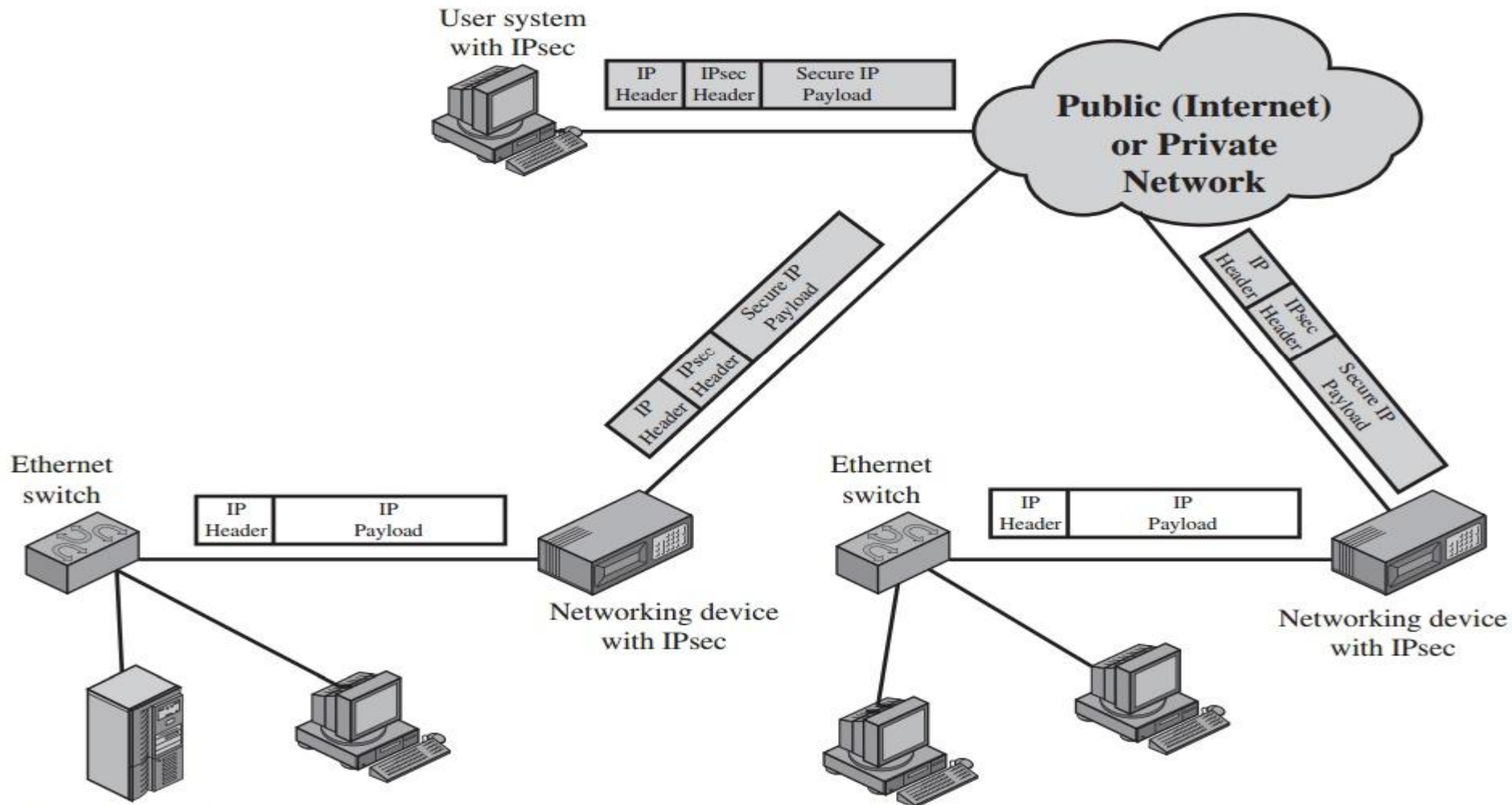


Figure 19.1 An IP Security Scenario

How IPSec Provides Secure Communication

- IPSec provides secure communication by using a combination of
 - ***Authentication***
 - ***Encryption protocols.***
- By authentication, the user is confident that no one else can access or intercept any data that is sent between two devices.
- Encryption also scrambles data into an unintelligible format so that only authorized users can have access of it.

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IP Security

Benefits of IP Security

- When IPSec is implemented in a firewall or router, it provides strong security for all traffic crossing
- IPSec is below the transport layer(TCP,UDP), so it is transparent to applications
- IPSec can be transparent to end users. There is no need to train the users on security mechanisms.
- IPSec can provide security for individual users if needed. This is useful for offsite workers for setting up a secure virtual subnetwork within an organization for sensitive applications.

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IP Security

Which protocol is commonly used to create secure VPN tunnels?

- A) HTTP
- B) FTP
- C) IPsec
- D) SMTP

Answer: C) Ipsec

What is the main purpose of IP Security (IPSec)?

- A) To manage domain names
- B) To provide secure communication over IP networks
- C) To compress data packets
- D) To control routing paths

Answer: B) To provide secure communication over IP networks

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IP Security

Which two protocols are the main components of IPSec?

- A) HTTP and HTTPS
- B) AH and ESP
- C) TCP and UDP
- D) FTP and SMTP

Answer: B) AH and ESP

What does the Authentication Header (AH) in IPSec provide?

- A) Data encryption only
- B) Authentication and integrity
- C) Compression and routing
- D) Error detection

Answer: B) Authentication and integrity

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IP Security

What does the Encapsulating Security Payload (ESP) in IPSec provide?

- A) Only authentication
- B) Only encryption
- C) Encryption, authentication, and integrity
- D) Only error control

Answer: C) Encryption, authentication, and integrity

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Dynamic Host Configuration Protocol



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Dynamic Host Configuration Protocol

How does the host get IP address?

- The large organizations or an ISP can receive a block of addresses directly from ICANN and small organizations can receive a block of addresses from an ISP.
- After a block of addresses are assigned to an organization, the network administration can manually assign addresses to the individual hosts or routers
- The **address assignment** in an organization can be done **automatically using the Dynamic Host Configuration Protocol (DHCP)**

(**ICANN** - The Internet Corporation for Assigned Names and Numbers)

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Dynamic Host Configuration Protocol

- DHCP is an **application-layer program**, using the client-server paradigm.
- It is often called as **plug and play protocol or zero conf protocol**.

Goal: allow host to **dynamically obtain its IP address from network server** when it joins the network

- Address is allocated on **leased basis**
- can **renew its lease** time to time
- allows **reuse of addresses** (only hold address while connected “on”)
- support mobile users who want to join network . Each time the user get connected to the network, it may get new address

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DHCP – Operations

DHCP Working:

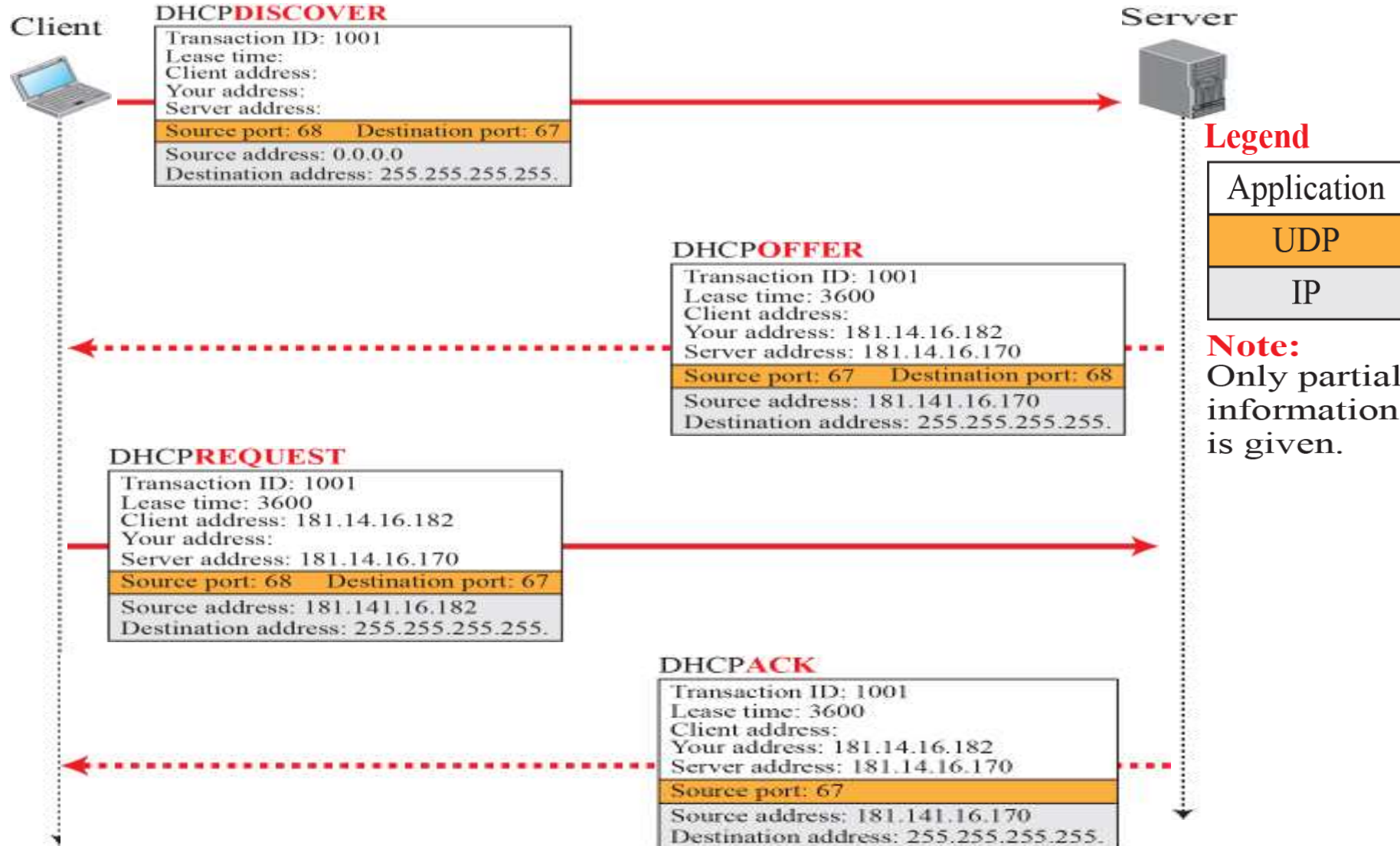
- host broadcasts “**DHCP discover**” msg (includes Transaction ID)
 - DHCP server responds with “**DHCP offer**” msg (broadcast)
 - host select the best of them and sends the request message : “**DHCP request**” msg (broadcast)
 - Selected DHCP server sends address: “**DHCP ack**” msg otherwise it may send “**DHCP NAK**” (broadcast)
-
- DHCP uses two **well known ports – 68(client) and 67 (server)**
 - DHCP is an application layer protocol **encapsulated in UDP datagram**

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DHCP operations

IP Address: ?

IP Address: 181.14.16.170



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DHCP message format

0	8	16	24	31
Opcode	Htype	HLen	HCount	
Transaction ID				
Time elapsed		Flags		
Client IP address				
Your IP address				
Server IP address				
Gateway IP address				
Client hardware address				
Server name				
Boot file name				
Options				

Fields:

Opcode: Operation code, request (1) or reply (2)

Htype: Hardware type (Ethernet, ...)

HLen: Length of hardware address

HCount: Maximum number of hops the packet can travel

Transaction ID: An integer set by client and repeated by the server

Time elapsed: The number of seconds since the client started to boot

Flags: First bit defines unicast (0) or multicast (1); other 15 bits not used

Client IP address: Set to 0 if the client does not know it

Your IP address: The client IP address sent by the server

Server IP address: A broadcast IP address if client does not know it

Gateway IP address: The address of default router

Server name: A 64-byte domain name of the server

Boot file name: A 128-byte file name holding extra information

Options: A 64-byte field with dual purpose described in text

DATA COMMUNICATIONS

DHCP – MCQ



1. What is the main function of DHCP?

- A) To assign IP addresses automatically to devices in a network
- B) To translate domain names to IP addresses
- C) To encrypt data before transmission
- D) To route packets between networks

Answer: A) To assign IP addresses automatically to devices in a network

2. Which of the following ports does DHCP use?

- A) TCP port 80
- B) UDP ports 67 and 68
- C) TCP port 25
- D) UDP port 53

Answer: B) UDP ports 67 and 68

DATA COMMUNICATIONS

DHCP – MCQ

3. What does DHCP stand for?

- A) Data Host Control Protocol
- B) Dynamic Host Configuration Protocol
- C) Domain Host Connection Protocol
- D) Direct Host Communication Protocol

Answer: B) Dynamic Host Configuration Protocol

4. Which message is sent by a client to find available DHCP servers?

- A) DHCPDISCOVER
- B) DHCPOFFER
- C) DHCPREQUEST
- D) DHCPACK

Answer: A) DHCPDISCOVER

Data Communications

Routing Protocols - RIP

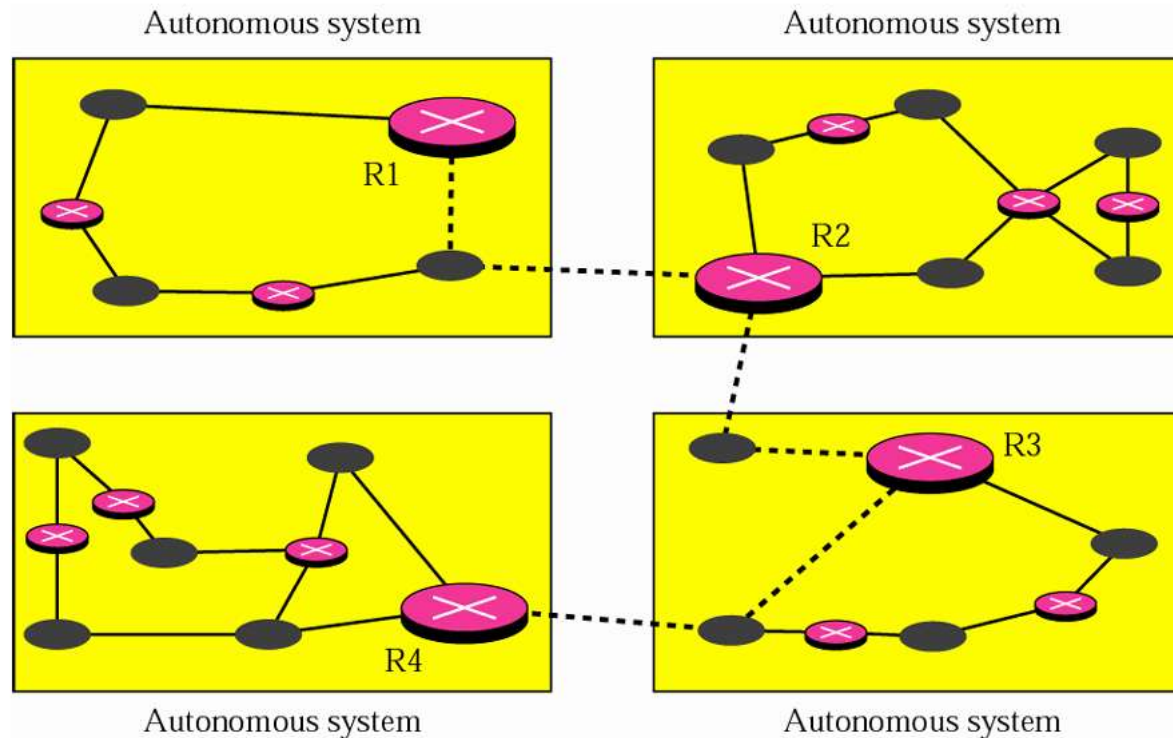


Routing Protocols

- Routing protocols are the set of rules and algorithms that routers use to communicate with each other to find the most efficient path to transmit data packets from a sender to a receiver.
- Routing is a procedure of moving data packets from one network to another by discovering the finest path from the source to the destination.
- The device that helps in finding the best path to forward data packets from one network to another network is called a **router**.

Routing Protocols

A group of networks and routers under authority of a single administrator is termed as an **Autonomous System**.



Routing Protocols

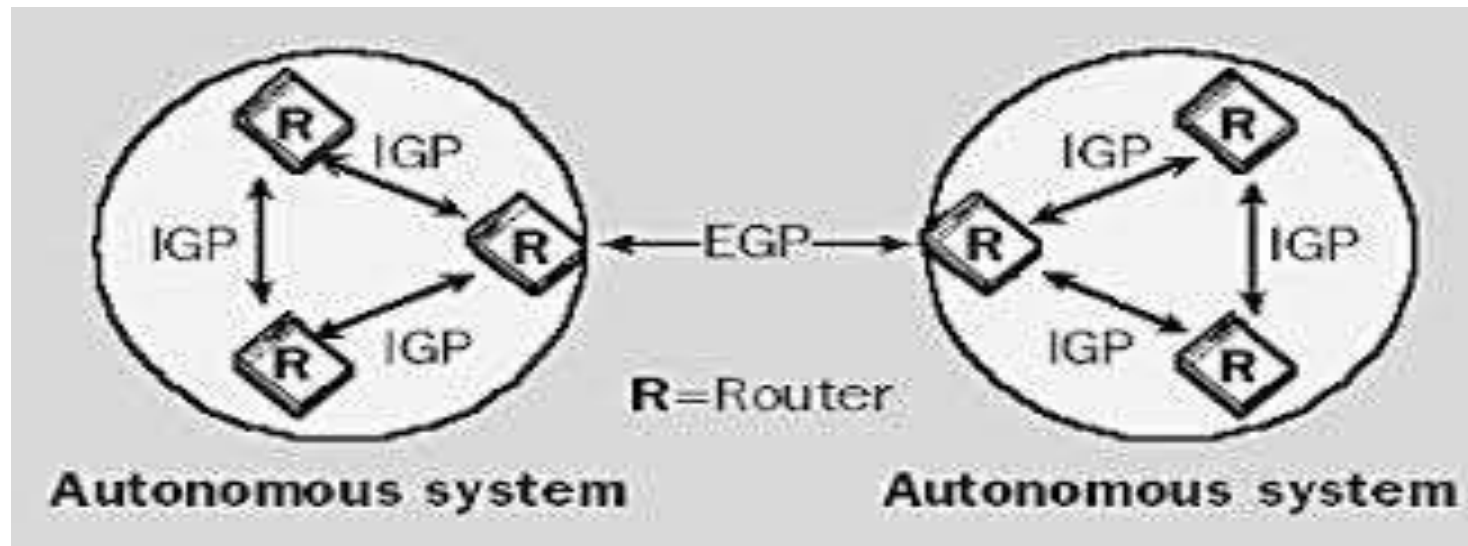
There are 2 types of routing protocols used in Autonomous system:

- 1. Interior gateway protocols (IGPs)**
- 2. Exterior gateway protocols (EGPs)**

Routing Protocols

Interior gateway protocols (IGPs):

- The routing protocols that are used within the same network or **autonomous system (AS)**.
- The main function of **IGP** is to search for the best route for forwarding data packets from sender to receiver within an autonomous system.

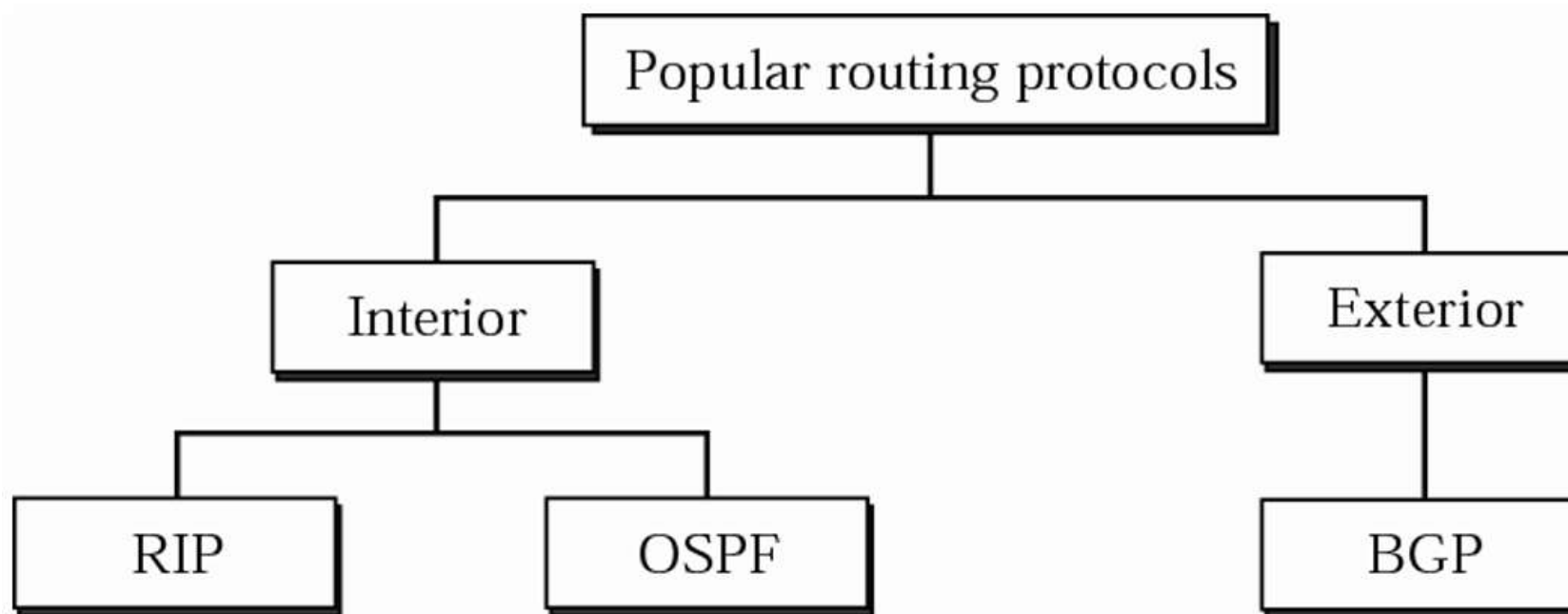


Exterior gateway protocols (EGPs):

- The routing protocols are used in various **autonomous systems** that facilitate the swapping of routing details between routers.
- The main function of **EGP** is to search for the best route for forwarding data packets from sender to receiver between various autonomous systems.

Routing Protocols

The main Routing protocols are classified as:



Routing Information Protocol (RIP)

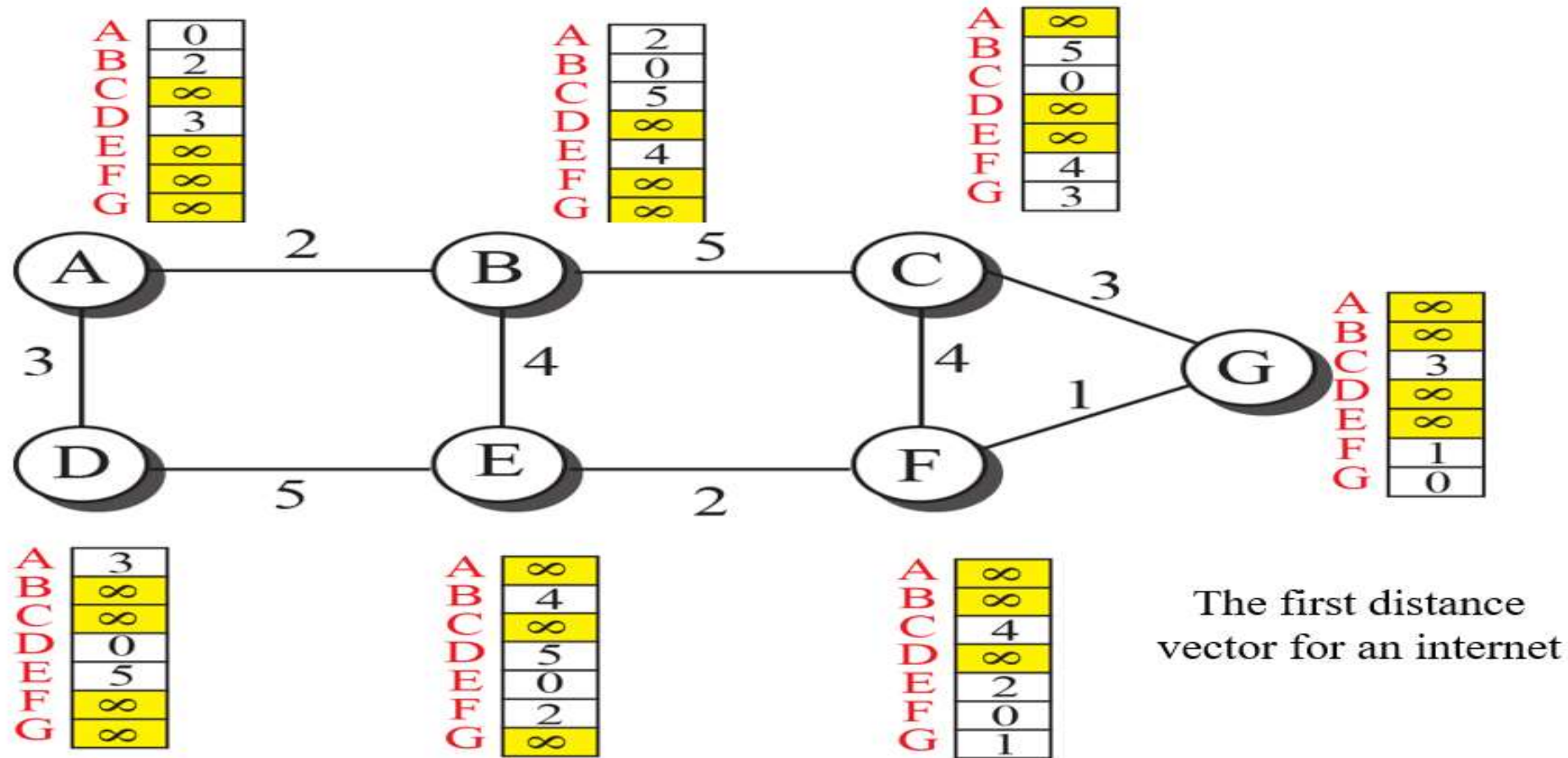
- RIP stands for Routing Information Protocol.
- RIP is an intra-domain routing protocol used within an autonomous system.
- RIP is based on the **distance vector-based strategy**, so consider the entire structure as a graph where nodes are the routers, and the links are the networks.
- In a routing table, the first column is the destination, or it is a network address.

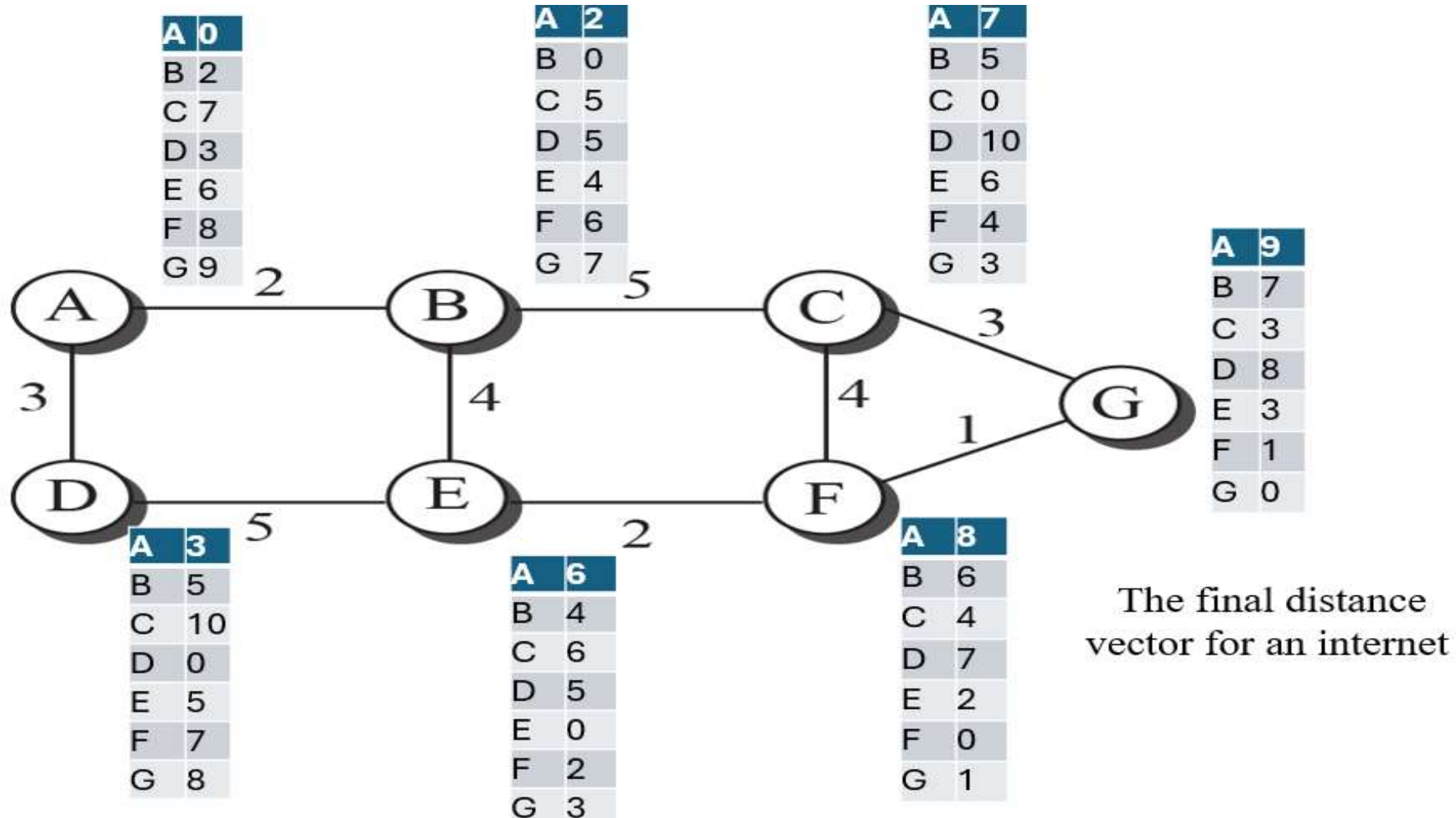
Routing Protocols - RIP

- The cost metric is the number of hops to reach the destination.
- The hop count is the number of networks required to reach the destination.
- In RIP, **infinity is defined as 16**, which means that the RIP is useful for smaller networks or small autonomous systems. The maximum number of hops that RIP can contain is 15 hops.
- The next column contains the address of the router to which the packet is to be sent to reach the destination.

Routing Protocols - RIP

- Each RIP router maintains a routing table.
- Each router broadcasts its routing table to its closest neighbors every 30 sec.
- The neighbors, in turn, pass the information on to their nearest neighbors, and so on, until all RIP hosts within the network have the same knowledge of routing paths. This shared knowledge is known as **convergence**.
- If a router receives an update on a route, and the new path is shorter, it will update its table entry with the length and next-hop address of the shorter path.
- If the new path is longer, it will wait through a "hold-down" period. It will only update the table entry if the new, longer path has been determined to be stable.





Data Communications

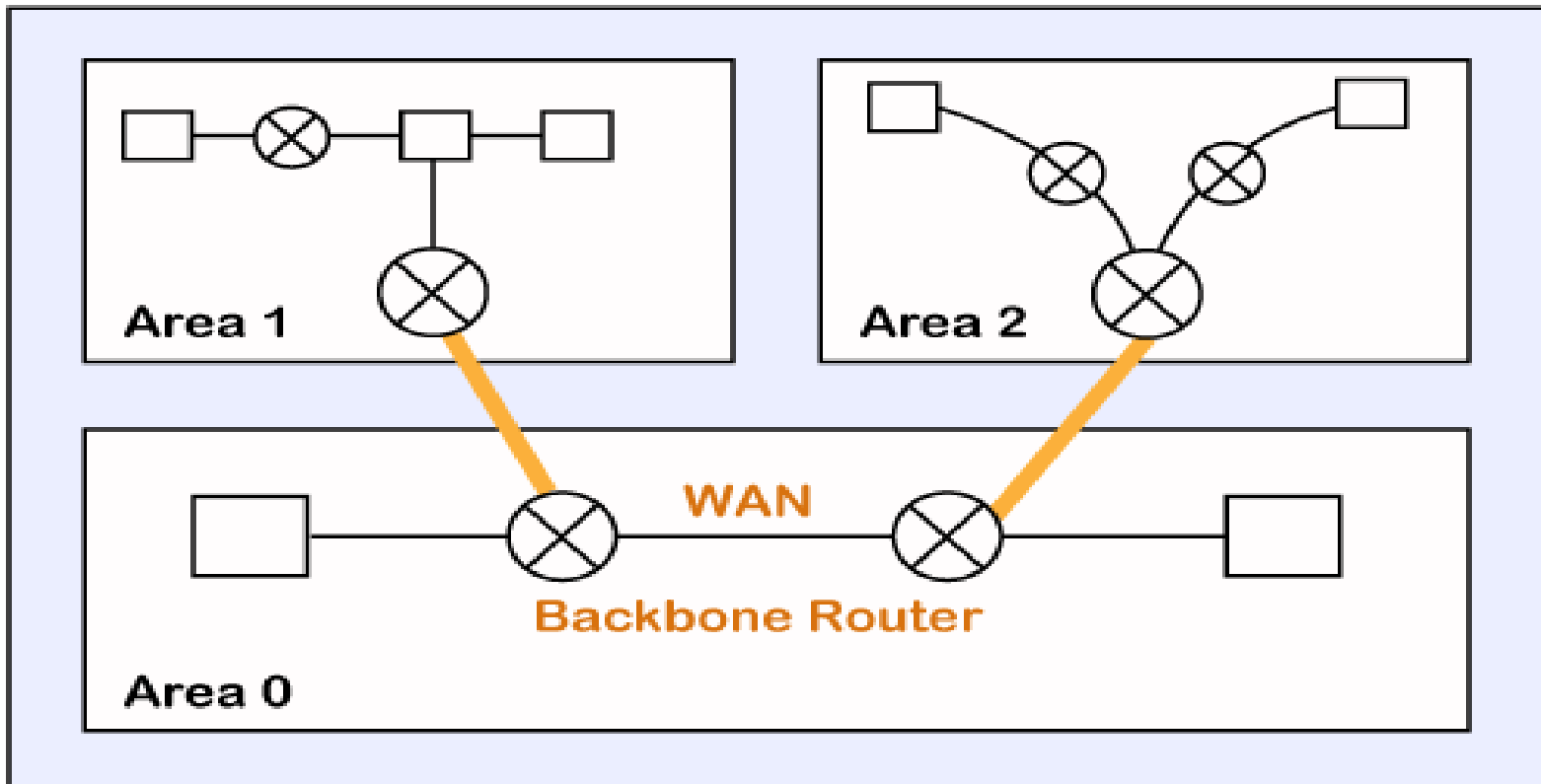
Routing Protocols - OSPF



Routing Protocols - OSPF

- The OSPF stands for **Open Shortest Path First**.
- It is an intradomain protocol, which is used within an area or a network.
- It is an interior gateway protocol, designed within a single autonomous system.
- It is based on a **link-state routing algorithm** in which each router contains the information of every domain, and based on this information, it determines the shortest path.
- The OSPF achieves by learning about every router and subnet within the entire network.

Fig: The OSPF Areas



Routing Protocols - OSPF

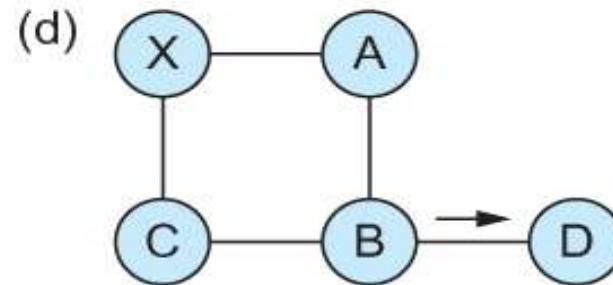
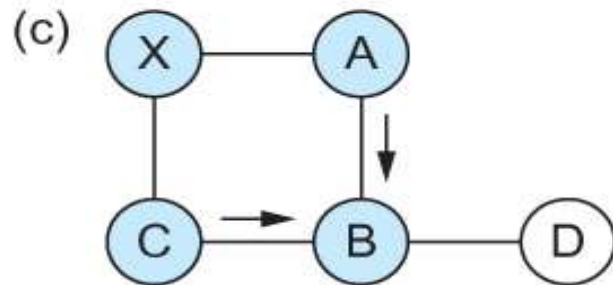
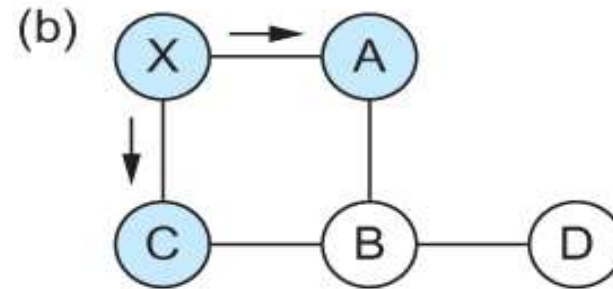
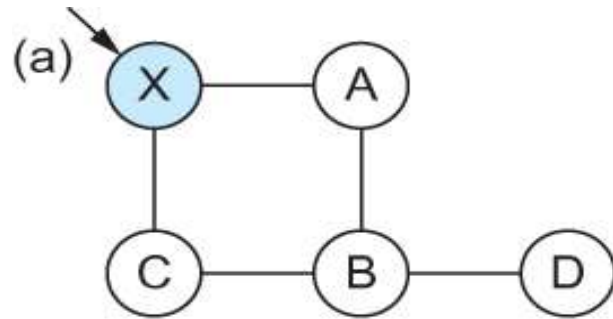
- OSPF divides the autonomous systems into Areas where the area is a collection of networks, hosts, and routers.
- Routers that exist inside the Area flood the area with routing information
- In Area, there is special router. These special routers are present at the border of an area, known as **Area Border Routers**. This router summarizes the information about an area and shares the information with other areas.

Routing Protocols - OSPF

- Every router learns the information about the network by **flooding** the **LSA (Link State Advertisements)**.
- Once the LSAs have been flooded, the OSPF stores the information in a link-state database known as **LSDB(Link state Database)**
- The main goal is to have the same information about every router in an LSDBs.

- **Link State Packet (LSP)**
 1. id of the node that created the LSP
 2. cost of link to each directly connected neighbor
 3. sequence number (SEQNO)
 4. time-to-live (TTL) for this packet

- 1 and 2 used for route calculation and
- 3 and 4 used for reliable flooding



Flooding of link-state packets.

(a) LSP arrives at node X;

(b) X floods LSP to A and C;

(c) A and C flood LSP to B (but not X);

(d) flooding is complete



THANK YOU

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